Assignment # 3
Process Synchronization: Producer–Consumer Problem Simulation

Due Date: December 31, 2025

Details

Students are asked to simulate a system where:

- Producers create items (e.g., integers or messages) and place them into a shared buffer.
- Consumers take items from the buffer for processing.
- The buffer has a limited capacity, and both producers and consumers must synchronize to avoid race conditions.

The simulation should:

1. Use threads or processes for producers and consumers.
2. Protect the shared buffer using:
   - Semaphores, locks, or synchronized blocks (depending on language).
3. Show real-time output of items being produced and consumed.
4. Prevent overfilling or empty consumption of the buffer.

Deliverables

Submit the following:

1. `source` — Actual code in Python/C/Java
2. Screenshot of the execution of the program capture
3. A `report.pdf` file containing all the steps you have performed and an explanation of the code.